

Baseline_gradient_steps:critic(policy evaluation or value evaluation)时用同一组数据运用 gradient-descent 训练 critic 内部神经网络

参数, 即找寻 $((\text{values} - \text{q_values})^{**2}).\text{mean}()$ 最小值时进行内部迭代的次数

Baseline_learning_rate:critic 内部迭代时的步长, 即每次向最优神经网络参数靠近时改变的速率

batch_size:每次外部迭代时, 采集数据的 state-action pairs 的数量

Discount:计算 state_value action_value 时的 gamma 值

Env_name:environment name 即该强化学习所处环境

Ep_len:episode length,一条轨迹的长度

Eval_batch_size:evaluation batch size, 即进行一次 evaluation 时使用多少 state-action pairs

Exp_name:experience name

Gae_lambda:gae 算法中将走 1, 2, 3.....n 步的 advantage (即残差) 进行几何加权, 得到 $A(\text{gae}), \text{gae} + \lambda$ 就是进行几何加权时的

参数 λ , 即 $A_t = \delta_t + \gamma \lambda \delta_{t+1} + (\gamma \lambda)^2 \delta_{t+2} + \dots$ 中的 λ

Layer_size:神经网络隐藏层中神经元的数目

Learning_rate:使用 policy gradient 时, 神经网络参数改变的速率, 即 $\theta_{t+1} = \theta_t - \alpha \nabla_{\theta} J(\theta)$ 中的 α

N_iter:number of iteration 外层迭代的次数

N_layers:number of layers, 神经网络隐藏的层数, 即为神经网络的深度

no_gpu:若为 true 则强制使用 cpu 跑, false 可用 gpu

Normalize_advantage:控制要不要对优势做标准化 (教程里没有这个, 没太明白说实话)

scalar_log_freq:记录日志的频率, 即搁多久迭代才往日志里写一次数据

Seed: 随机种子, 一个整数, 用来固定所有随机性, 让每次运行结果可复现。

Use_baseline:是否使用 baseline

Use_reward_to_go:是否使用 rewardto go, 即从当前 t 开始记录 reward, 不考虑之前的数

Video_log_freq:记录视频频率

Which_gpu:用哪块 gpu 配置项

Train stdreturn: 训练时 retrun 的标准差

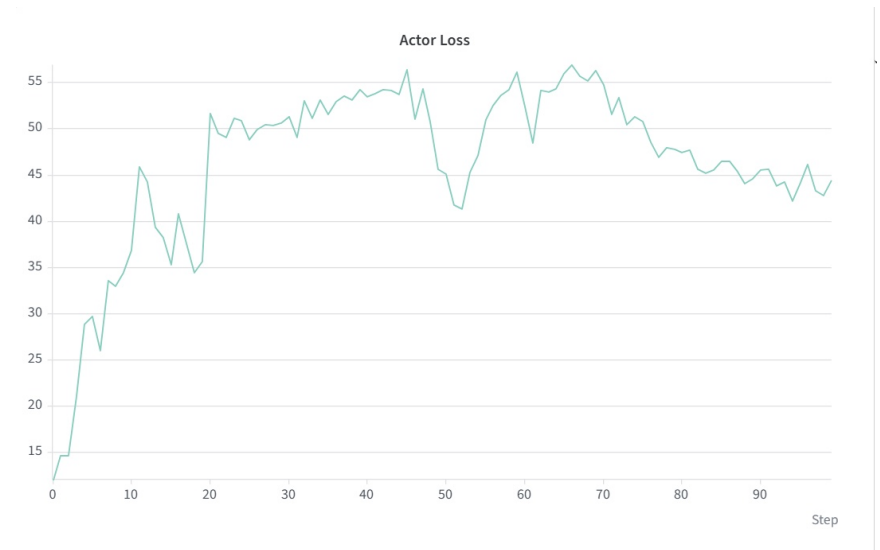
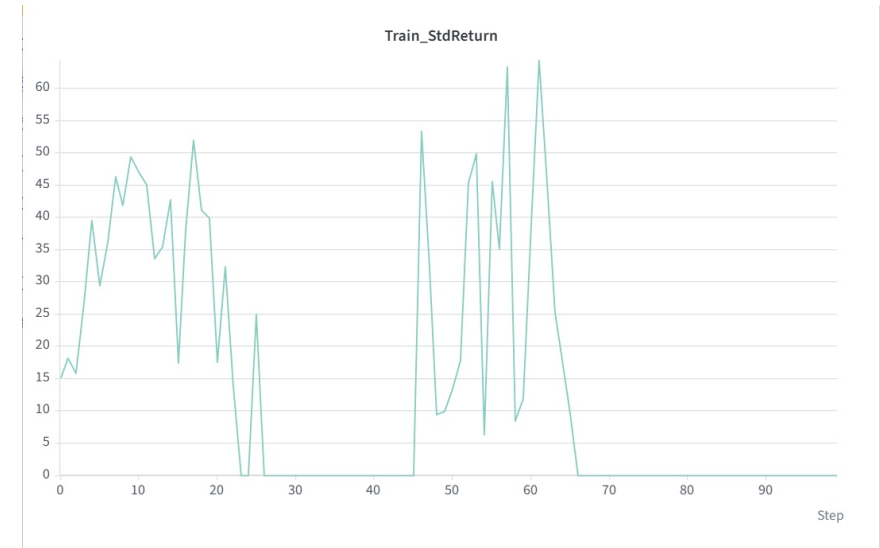
Actor loss: 策略网络遗失, 为使好动作的概率变大, 坏概率的动作变小

Baseline loss: 显示预测的 statue value 和实际的 action value 之间的差, 越小 advantage 越好

—、vanilla policy gradient

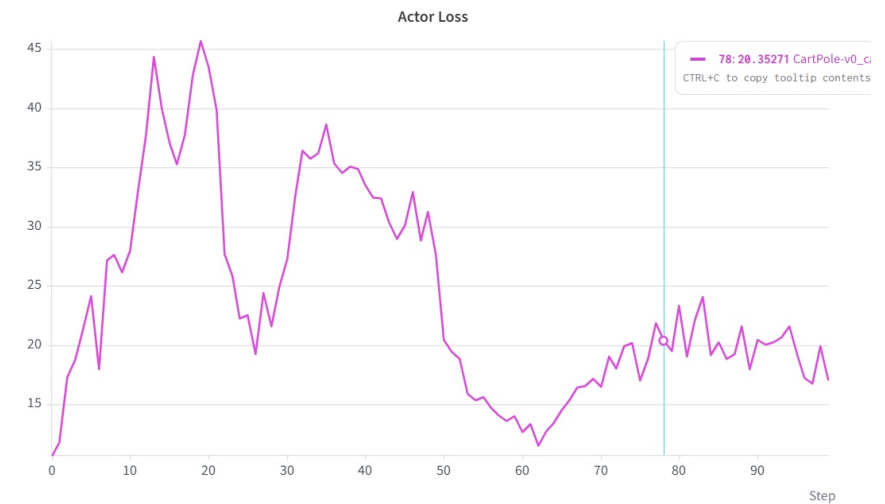
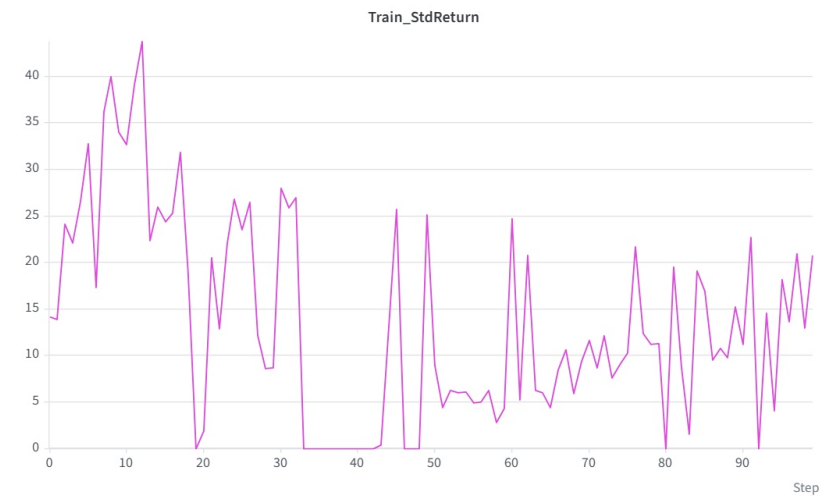
1. Cartpole

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:1000
Discount:1
Env_name:CartPole-v0
Ep_len:null
Eval_batch_size:400
Exp_name:cartpole
Gae_lambda:null
Layer_size:64
Learning_rate:0.005
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1
Use_baseline:false
Use_reward_to_go:false
Video_log_freq:-1
Which_gpu:0



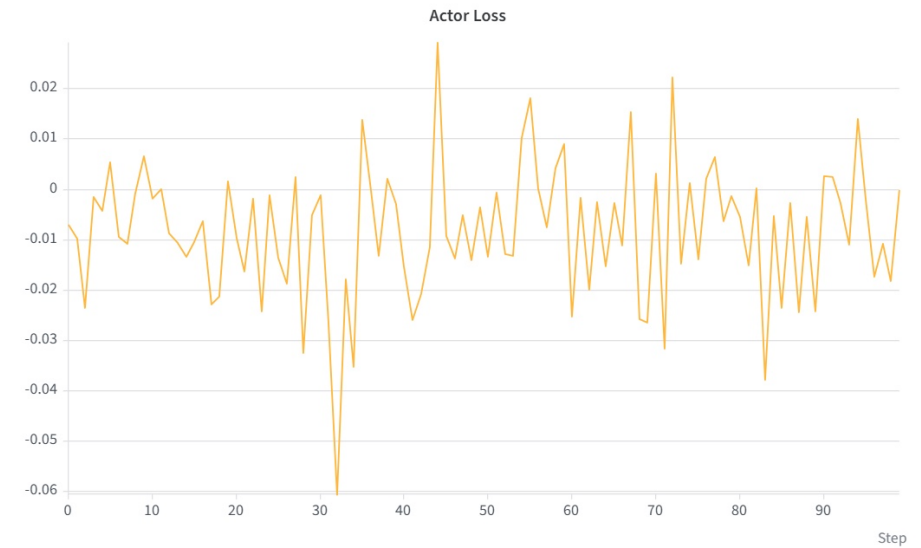
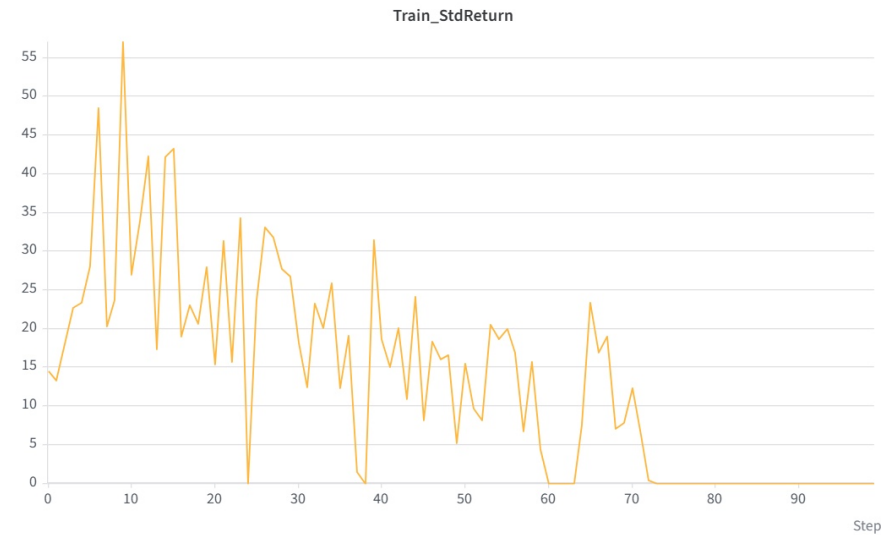
2. Cartpole_rtg

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:1000
Discount:1
Env_name:CartPole-v0
Ep_len:null
Eval_batch_size:400
Exp_name:cartpole_rtg
Gae_lambda:null
Layer_size:64
Learning_rate:0.005
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1
Use_baseline:false
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0



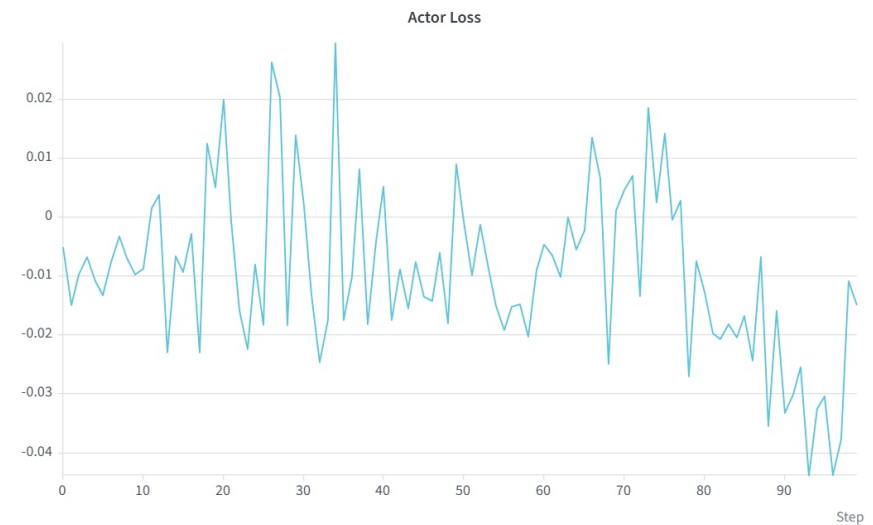
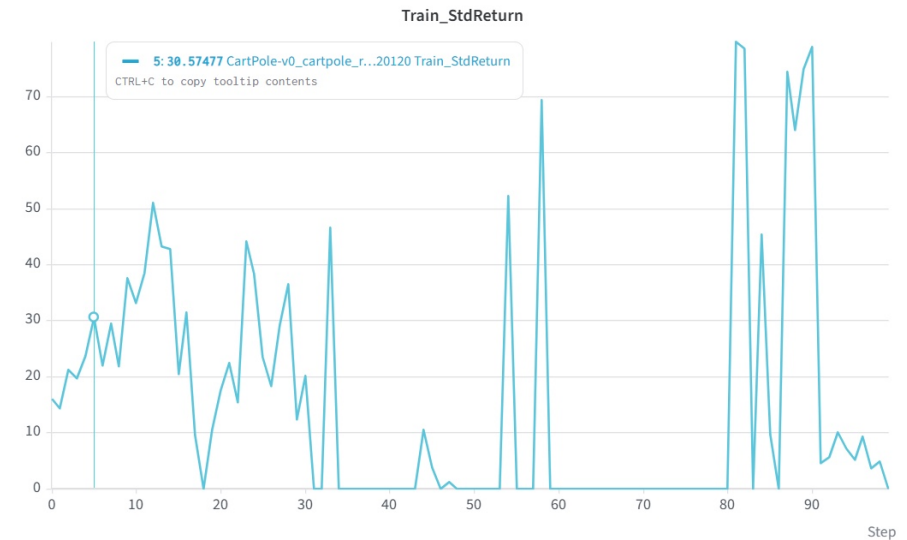
3. Cartpole_na

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:1000
Discount:1
Env_name:CartPole-v0
Ep_len:null
Eval_batch_size:400
Exp_name:cartpole_na
Gae_lambda:null
Layer_size:64
Learning_rate:0.005
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:true
scalar_log_freq:1
Seed: 1
Use_baseline:false
Use_reward_to_go:false
Video_log_freq:-1
Which_gpu:0



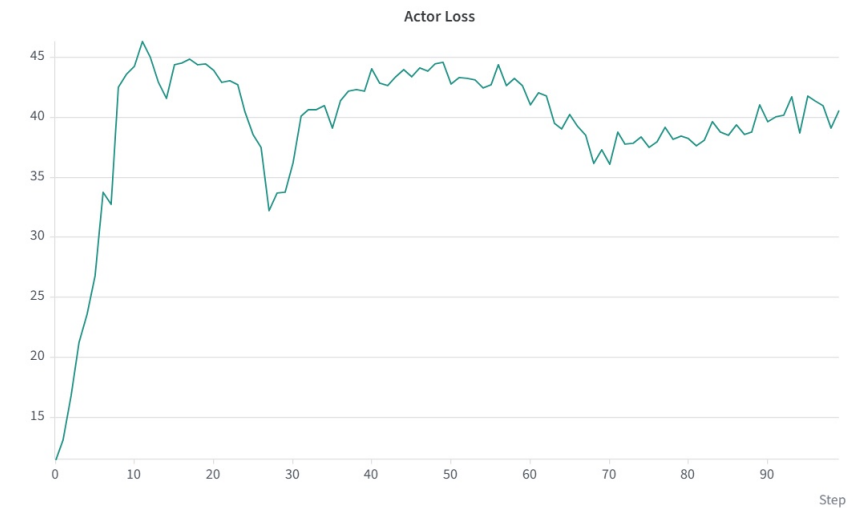
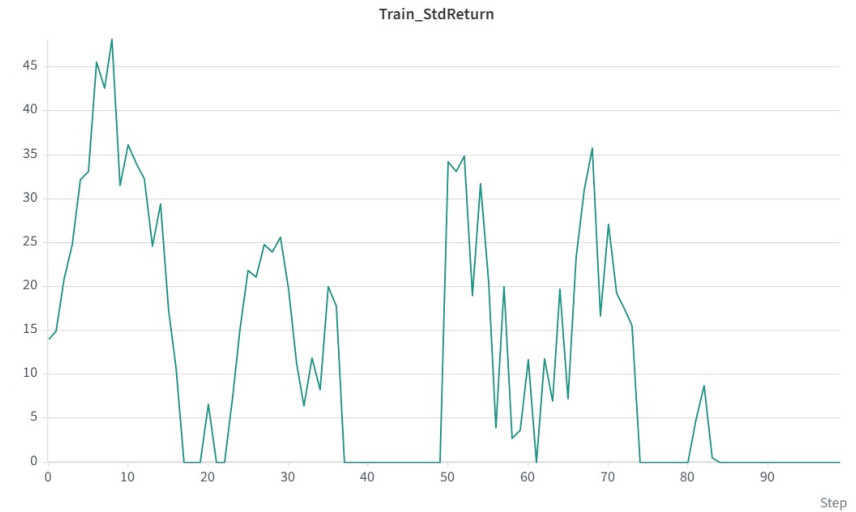
4. Cartpole_rtg_na

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:1000
Discount:1
Env_name:CartPole-v0
Ep_len:null
Eval_batch_size:400
Exp_name:cartpole_rtg_na
Gae_lambda:null
Layer_size:64
Learning_rate:0.005
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:true
scalar_log_freq:1
Seed: 1
Use_baseline:false
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu



5. Cartpole_lb

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:4000
Discount:1
Env_name:CartPole-v0
Ep_len:null
Eval_batch_size:400
Exp_name:cartpole_lb
Gae_lambda:null
Layer_size:64
Learning_rate:0.005
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1
Use_baseline:false
Use_reward_to_go:false
Video_log_freq:-1
Which_gpu:0



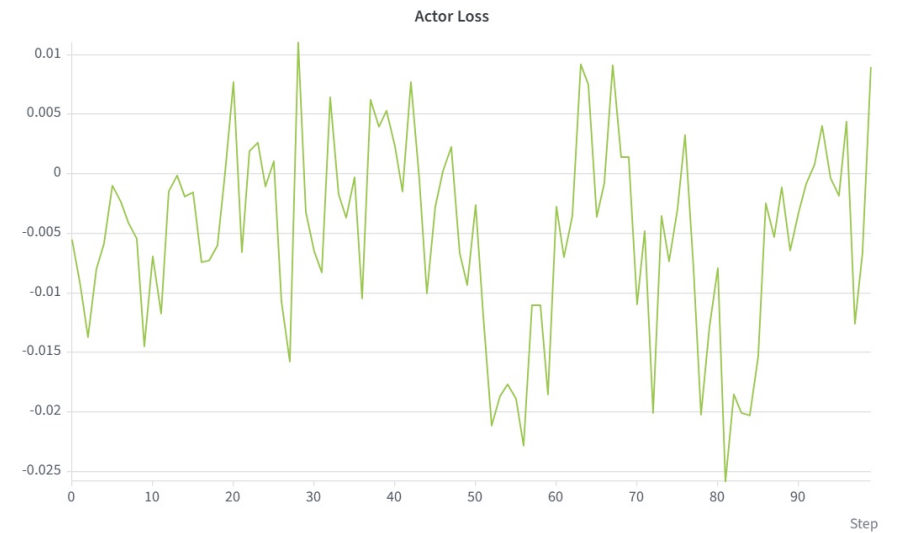
6. Cartpole_lb_rtg

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:4000
Discount:1
Env_name:CartPole-v0
Ep_len:null
Eval_batch_size:400
Exp_name:cartpole_lb_rtg
Gae_lambda:null
Layer_size:64
Learning_rate:0.005
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1
Use_baseline:false
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0



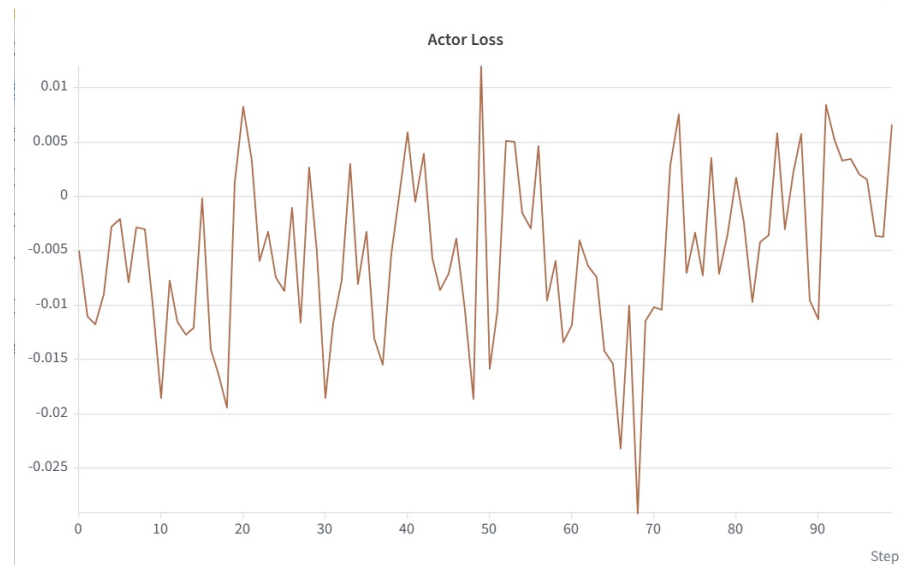
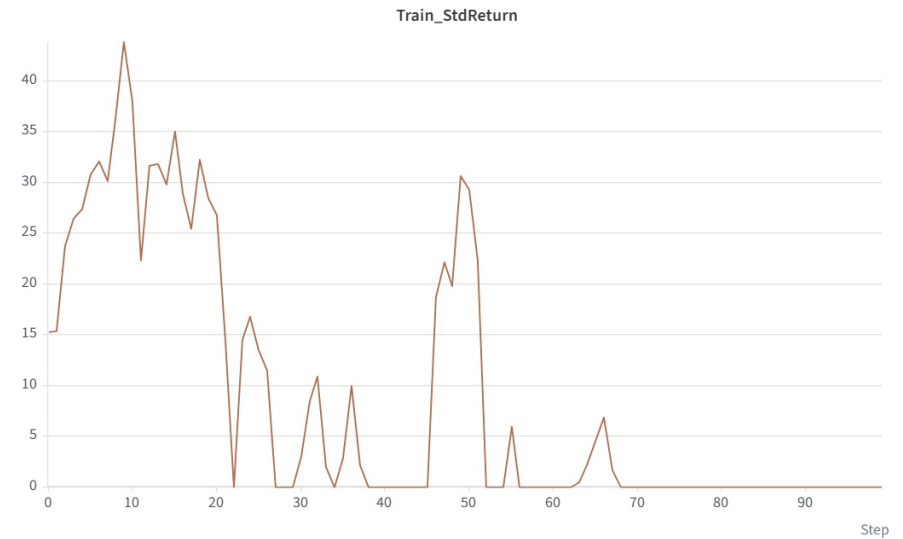
7. Cartpole_lb_na

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:4000
Discount:1
Env_name:CartPole-v0
Ep_len:null
Eval_batch_size:400
Exp_name:cartpole_lb_na
Gae_lambda:null
Layer_size:64
Learning_rate:0.005
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:true
scalar_log_freq:1
Seed: 1
Use_baseline:false
Use_reward_to_go:false
Video_log_freq:-1
Which_gpu:0



8. Cartpole_lb_rtg_na

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:4000
Discount:1
Env_name:"CartPole-v0"
Ep_len:null
Eval_batch_size:400
Exp_name:"cartpole_lb_rtg_na"
Gae_lambda:null
Layer_size:64
Learning_rate:0.005
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:true
scalar_log_freq:1
Seed: 1
Use_baseline:false
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0

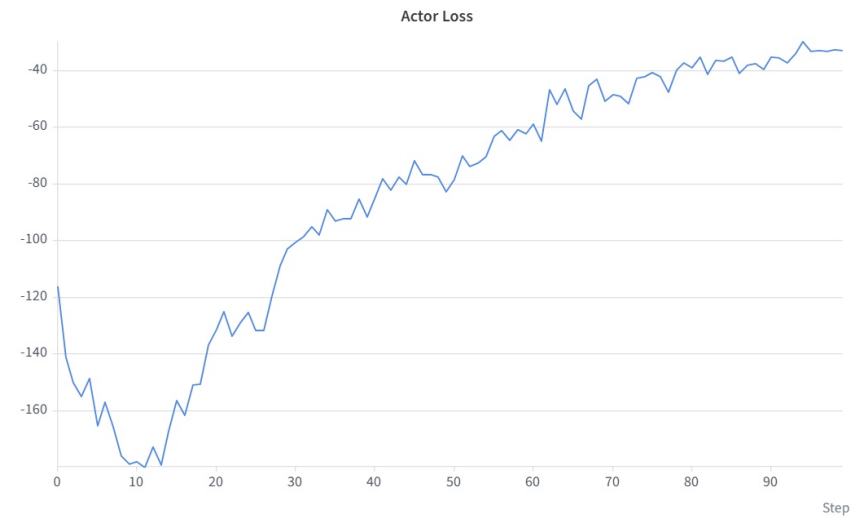
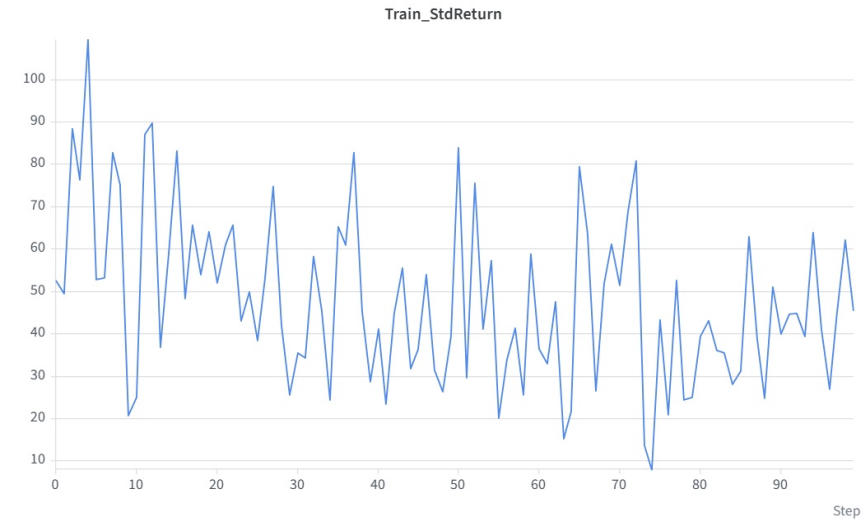


二、Using Baseline

1.cheetah

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:5000
Discount:0.95
Env_name:"HalfCheetah-v4"
Ep_len:null
Eval_batch_size:3000
Exp_name:"cheetah"
Gae_lambda:null
Layer_size:64
Learning_rate:0.01
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1

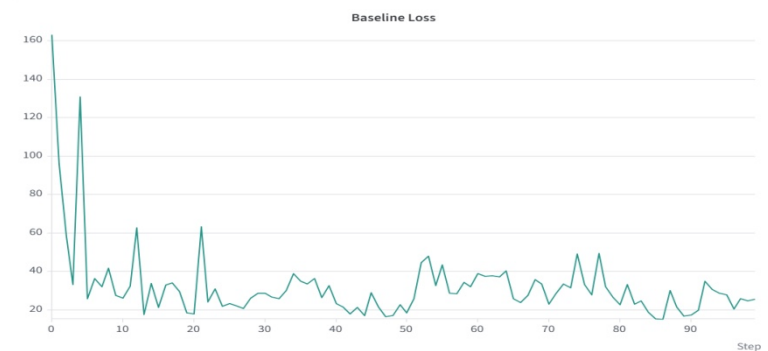
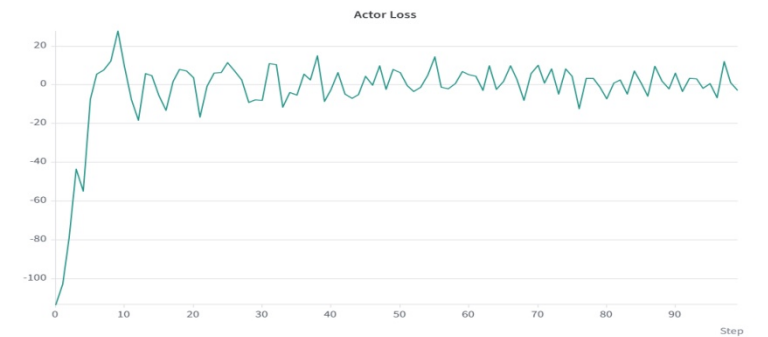
Use_baseline:false
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0



2. cheetah_baseline

Baseline_gradient_steps:5
Baseline_learning_rate:0.01
batch_size:5000
Discount:0.95
Env_name:"HalfCheetah-v4"
Ep_len:null
Eval_batch_size:3000
Exp_name:"cheetah_baseline"
Gae_lambda:null
Layer_size:64
Learning_rate:0.01
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1

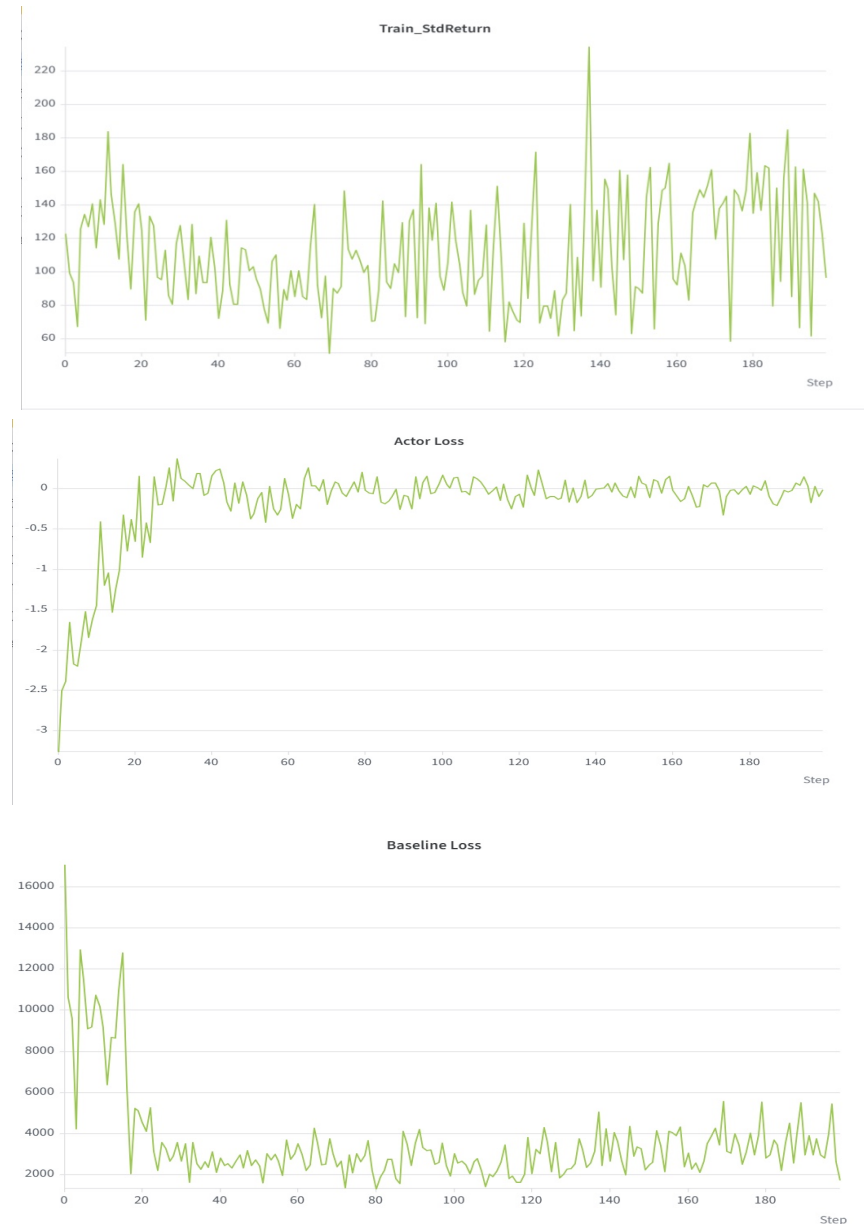
Use_baseline:true
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0



三、Implementing GAE

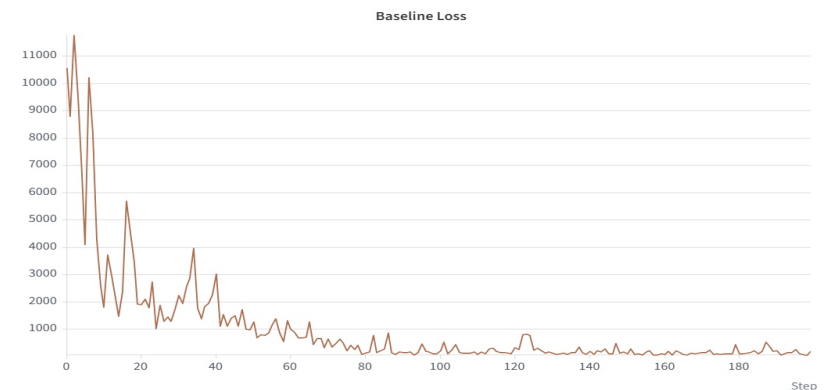
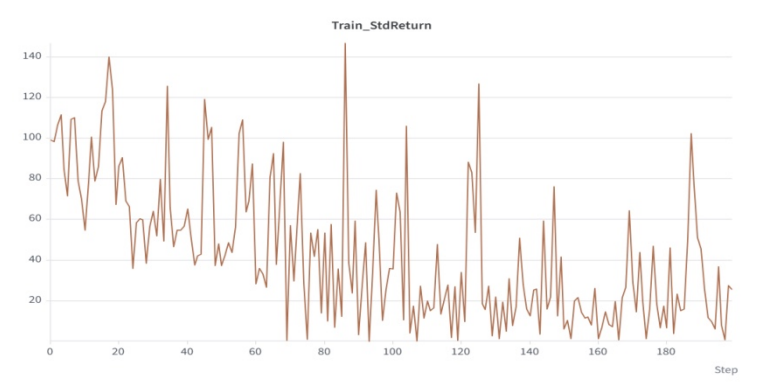
1.lunar_lander_lambda0

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:2000
Discount:0.99
Env_name:"LunarLander-v2"
Ep_len:1000
Eval_batch_size:2000
Exp_name:"lunar_lander_lambda0"
Gae_lambda:0
Layer_size:128
Learning_rate:0.001
N_iter:200
N_layers:3
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1
Use_baseline:true
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0



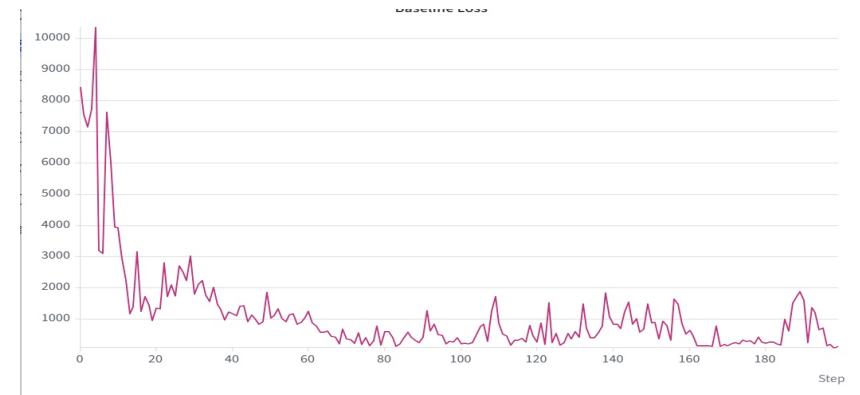
2.lunar_lander_lambda0.95

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:2000
Discount:0.99
Env_name:"LunarLander-v2"
Ep_len:1000
Eval_batch_size:2000
Exp_name:"lunar_lander_lambda0.95"
Gae_lambda:0.95
Layer_size:128
Learning_rate:0.001
N_iter:200
N_layers:3
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1
Use_baseline:true
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0



3.lunar_lander_lambda0.98

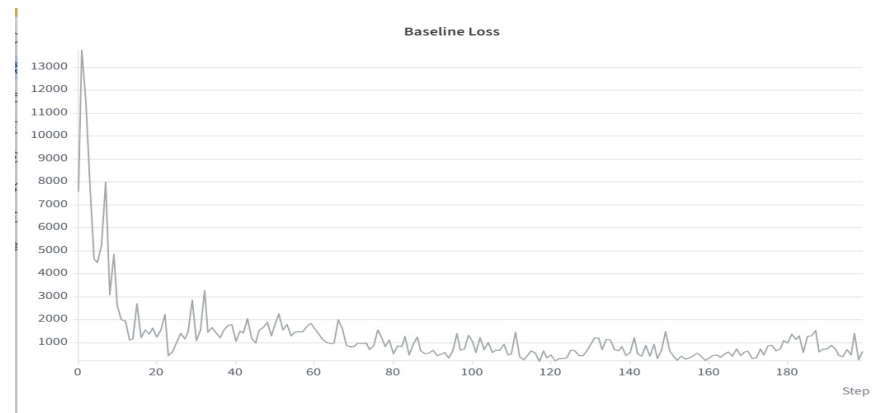
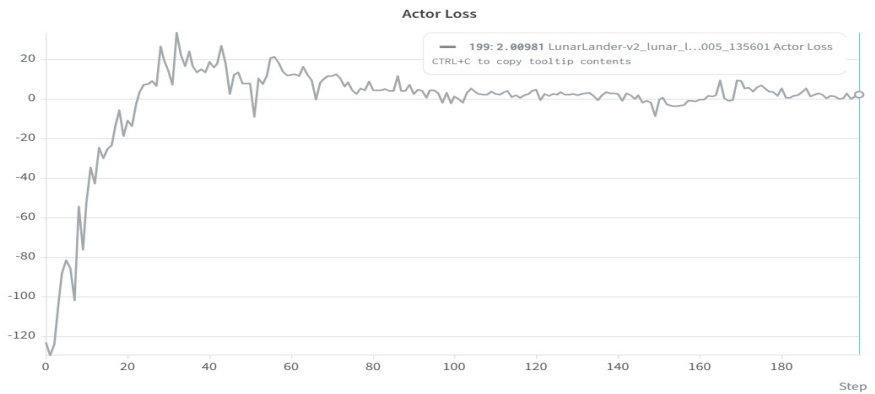
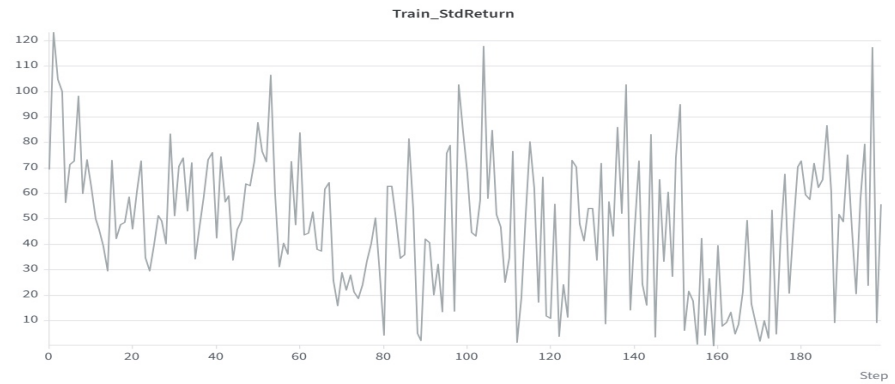
Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:2000
Discount:0.99
Env_name:"LunarLander-v2"
Ep_len:1000
Eval_batch_size:2000
Exp_name:"lunar_lander_lambda0.98"
Gae_lambda:0.98
Layer_size:128
Learning_rate:0.001
N_iter:200
N_layers:3
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1
Use_baseline:true
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0



4.lunar_lander_lambda0.99

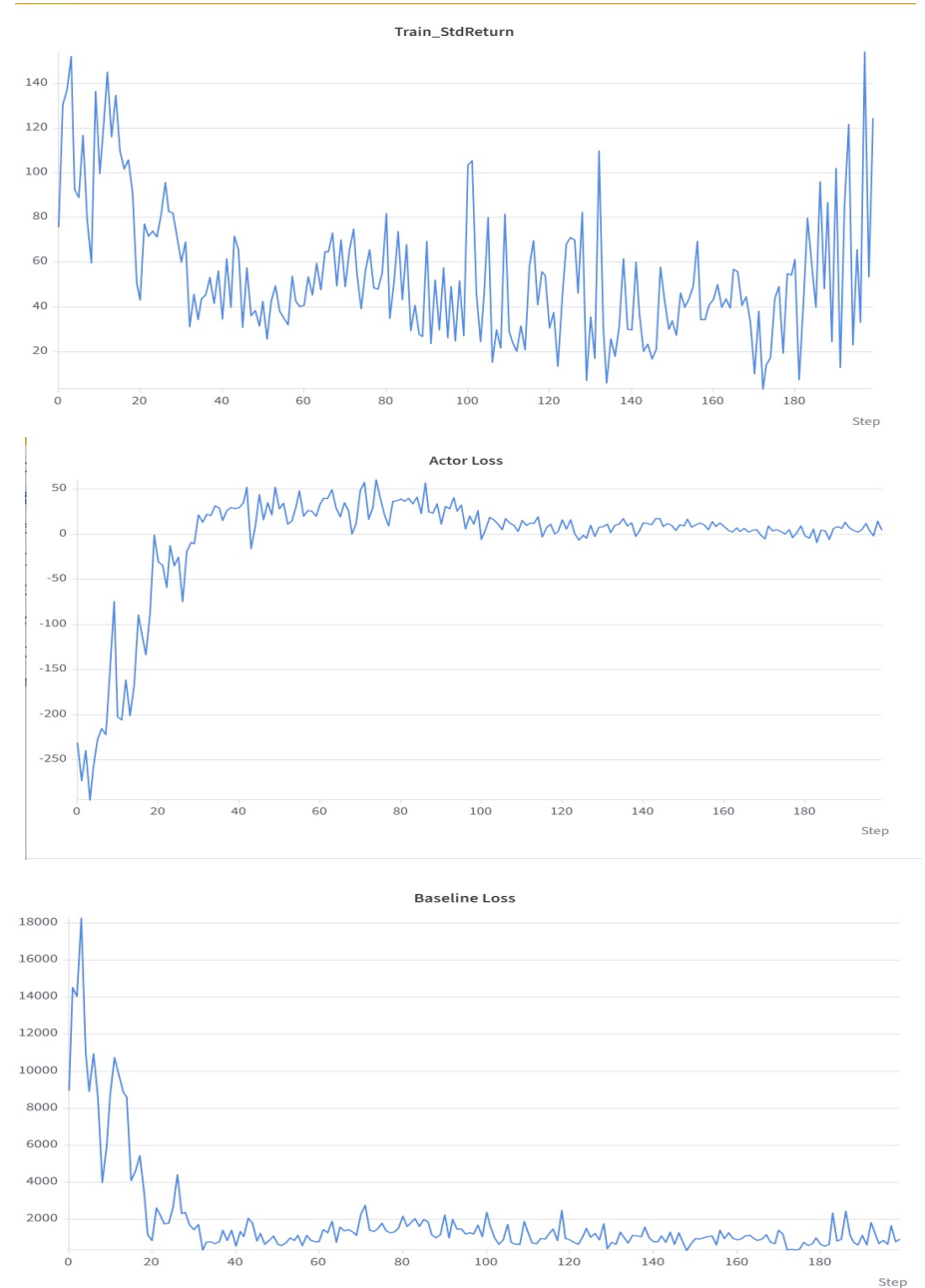
Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:2000
Discount:0.99
Env_name:"LunarLander-v2"
Ep_len:1000
Eval_batch_size:2000
Exp_name:"lunar_lander_lambda0.99"
Gae_lambda:0.99
Layer_size:128
Learning_rate:0.001
N_iter:200
N_layers:3
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1

Use_baseline:true
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0



5. lunar_lander_lambda1

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:2000
Discount:0.99
Env_name:"LunarLander-v2"
Ep_len:1000
Eval_batch_size:2000
Exp_name:"lunar_lander_lambda1"
Gae_lambda:1
Layer_size:128
Learning_rate:0.001
N_iter:200
N_layers:3
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1
Use_baseline:true
Use_reward_to_go:true
Video_log_freq:-1
Which_gpu:0



四、Hyperparameters and Sample Efficiency

Baseline_gradient_steps:5
Baseline_learning_rate:0.005
batch_size:5000
Discount:1
Env_name:"InvertedPendulum-v4"
Ep_len:null
Eval_batch_size:1000
Exp_name:"pendulum"
Gae_lambda:null
Layer_size:64
Learning_rate:0.005
N_iter:100
N_layers:2
no_gpu:false
Normalize_advantage:false
scalar_log_freq:1
Seed: 1
Use_baseline:false
Use_reward_to_go:false
Video_log_freq:-1
Which_gpu:0

